Disparity Analyses Are Robust To Ancestral State Estimation Uncertainty

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Abstract

Ancestral state estimation is widely used to infer unobserved traits across a phylogeny, extending analyses beyond incomplete fossil records. In studies of morphological disparity ancestral state estimation can account for ghost lineages, yet estimates are often confounded with high uncertainty. How best to deal with this uncertainty remains unclear. Using simulations, we compared pre- and post-ordination estimation methods, and both point estimate and probabilistic approaches for treating uncertainty, under continuous and discrete trait models and varying fossil sampling levels. The relative disparity error was calculated from the difference between our estimated disparity values and the true disparity values from the original simulations. We then extended this framework to evaluate the recovery of changes in disparity across time by simulating mass extinction events with varying intensity and selectivity. We find that pre-ordination probabilistic estimations outperform post-ordination and point estimation methods and that omission of ancestral state estimations leads to underestimates of disparity. As expected, higher fossil sampling improves estimation accuracy, although recovered disparity accuracy was robust even at low fossil sampling. Similar patterns were observed when recovering changes in disparity across mass extinction events but declined significantly when fossil sampling was low, highlighting that fossil inclusion is essential when assessing disparity in deep time. These results demonstrate that explicitly propagating uncertainty, rather than collapsing states into point estimates, is more likely to recover accurate patterns in disparity, reinforcing the need to move beyond point estimates when working with the incomplete fossil record.

Keywords: ancestral state estimation; morphological disparity; fossils; simulation; uncertainty; macroevolution

1 Introduction

2 Materials and Methods

2.1 Simulation Framework

2.1.1 Phylogenies

All analyses were carried out using R v 4.5.2 (R Core Team 2025). We simulated birth-death trees with the speciation (λ) parameter set to 1.0, and extinction (μ) set to 0.9 (Guillerme 2024). Fixing $\lambda = 1.0$ provides a convenient relative time-scale, since only the ratio of extinction to speciation affects the shape and balance of simulated trees. These parameters correspond to a high turnover regime, consistent with the upper-empirical estimates from diversification studies of animal groups (Scholl and Wiens 2016; Pires et al. 2018; Henao Diaz et al. 2019; Morlon et al. 2024). This level of extinction produces realistic evolutionary histories with substantial extant and extinct diversity, while maintaining a positive net diversification rate ($\lambda - \mu > 0$).

We generated three sets of 100 birth-death trees with these parameters until the simulations reached 50, 100 and 150 living taxa, with maximum tree size of 500, 1000 and 1500 tips, respectively. The randomness of the birth-death model meant some simulated trees contained only extinct taxa, so we re-ran each individual simulation until we obtained the desired number of living taxa (50, 100 or 150). This meant that although the parameters were set to $\lambda = 1$ and $\mu = 0.9$, the effective parameters slightly deviated from these input values (see Supplementary Figure 1). Mean net diversification across all simulated trees was 0.30 ± 0.09.

2.1.2 Trait simulations

For each phylogeny, we simulated continuous and discrete trait data separately. Continuous trait data were simulated under variations of Brownian motion (BM) and Ornstein-Uhlenbeck (OU) models (table 1). We simulated BM with variance = 0.25. We also simulated BM with a trend (BM_t), whereby the ancestral trait value increased by an increment of 0.1 multiplied by branch length. For OU models, the data were simulated with strong pull to the optimum (α) (OU_{st}), weak α (OU_w) and with differing optimum and root states (OU_{sh}). Following Gearty et al. (2024), the α parameter was set based on phylogenetic half life ($\frac{ln(2)}{\alpha}$) to account for differing tree heights across simulations (Cooper et al. 2016). For the strong and weak OU models, α was set as 25% and 75% of the tree height respectively. We simulated 20 independent characters for

each model (BM, BM_t, OU_w, OU_{st}, OU_{sh}) across each phylogeny.

Discrete trait evolution was simulated using a discrete-time Markov process, where the probability of state transition along each branch was proportional to the product of the transition probability and branch length. We simulated discrete traits across three rate categories, representing fast (0.9), medium (0.1) and slow (0.01) transition rates (table 1). These rates were assigned by generating equal-rates (ER) transition probability matrices. For each category, 85 binary characters and 15 three-state characters were simulated, following a typical scoring ratio found in empirical data (Guillerme and Cooper 2016; Casali et al. 2023).

2.2 Fossil sampling

To simulate different levels of fossil sampling, extinct tips were either retained ('fossilized') or pruned using a Bernoulli sampling method at five different probabilities: 0%, 5%, 15%, 50% and 100%. This method means each extinct tip had an equal probability of being 'fossilized' at each sample level, irrespective of the total tree size. The corresponding trait matrices were subset to match each downsampled tree.

2.3 Ancestral state estimation (ASE)

We ran ancestral state estimation on the down-sampled matrices using the `dispRity::multi.ace` (Guillerme 2018), which is a wrapper for `ape::ace` (Paradis and Schliep 2019). We used four combinations of two approaches to ancestral state estimation to evaluate each method's accuracy in recovering the true disparity signal. These were then compared to a null model bypassing ancestral state estimation (Fig. 2, Step 4).

2.3.1 Pre-ordination estimation

Ancestral state estimation of trait data can be carried out at two points either side of the ordination step in the disparity pipeline (Lloyd 2018). Pre-ordination estimation (pre-OASE) involves the estimation of the raw trait values for each node. Post-ordination estimation (post-OASE) calculates the continuous scores across each PC axis of an ordination matrix.

Pre-OASE of continuous characters models evolution under a Brownian motion model, where ancestral state values lie within a multivariate normal distribution that is estimated given the observed tip data and the shared evolutionary history among taxa (Schluter et al. 1997; Rohlf 2001). Confidence intervals were calculated from standard errors derived from the Hessian matrix

at the maximum likelihood estimate. The width of confidence intervals reflects the uncertainty in continuous ancestral estimates, which typically increases with node depth and decreases with the number of informative descendant lineages.

The pre-OASE method for estimating ancestral discrete characters involves estimating the marginal likelihoods for each discrete state at each node, based on a maximum likelihood derived estimate of the Mk transition matrix and the tree topology given (Pagel 1999; Yang 2006). Given that the transition rates were set *a priori* during simulations, the equal-rates (ER) model was used to avoid model misspecification as a confounding factor on estimation accuracy (Keating 2023).

We derived trait spaces from which to estimate disparity through ordination of these estimations. For discrete characters, distance matrices were computed using the Maximum Observed Rescaled Distance metric (MORD) (Lloyd 2016) and ordinated using principal coordinates analyses (PCoA) (`cmdscale()` function in R). Continuous data were ordinated using principal components analysis (PCA) (`prcomp()` function in R).

2.3.2 Post-ordination estimation

For post-OASE, the same ordination step was applied prior to ancestral state estimation to the down-sampled discrete and continuous character matrices to generate a set of tip-only trait spaces. Ancestral node states were then estimated under a Brownian motion model across all PC axes, as detailed in section 2.3.1.

2.3.3 Uncertainty treatment

For each estimation method, we also compared point estimate and probabilistic approaches. The point estimate approach returns a fixed single value for each estimate. For continuous characters, the point estimate returned was the median of the ancestral estimate confidence interval. For discrete characters, we used a strict majority rule where the state with the highest scaled likelihood was selected at each node for each character. In scenarios where state likelihoods were equal, one state was chosen at random. We also used a relative majority rule, where any state at a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states was collapsed, otherwise states were left as uncertain (supplementary).

The probabilistic approach differs in that uncertainty in estimates are explicitly accounted for by sampling multiple results rather than a single point estimate. For continuous characters,

we sampled each state across 100 matrices, with each estimate value lying within a normal distribution centered on the mean ancestral estimation and spread across the estimated confidence interval. Each discrete character state at each node was sampled 100 times according to their scaled likelihoods (resulting in 100 character estimation matrices). In a scenario where there is uncertainty in a binary character for a specific node, e.g., 0.55 for state 0 and 0.45 for state 1, this distribution approach results in a 55% probability of sampling state 0 in each matrix for the respective node (and 45% chance of sampling state 1). Both procedures were implemented using the `sample = 100` argument in the `dispRity::multi.ace` function (Guillerme 2018).

2.4 Disparity analyses

To ensure fair comparison among different fossil sampling levels, only the nodes present across all fossil sampling levels were retained from the ancestral state estimation outputs for each phylogeny; therefore, comparisons in disparity calculations among fossil sampling levels differed solely on the difference in number of fossil tips and the estimations of node states.

When assessing overall disparity, we wanted to understand how well ancestral state estimation can recover true patterns in size and density in trait space occupancy (Guillerme et al. 2020). Size-based metrics describe the spread of points in the trait space (Guillerme et al. 2020). To explore how well the size of the trait space was recovered, we calculated the sum of quantiles. The sum of quantiles calculates the range on each PC axis for the 95% of the data and then sums these ranges across all axes. This metric has similar properties to the often used sum of ranges metric by describing the total trait space exploited (Foote 1992), but with less sensitivity to outliers and sample size. For density, we used the sum of variances (Ciampaglio et al. 2001; Wills 2001) and the average nearest neighbour distance (Foote 1992). Disparity values were calculated across each ancestral state estimation and the true simulated values so that a relative disparity error could be calculated (Fig. 2 Step 5).

$$\text{Relative disparity error} = \frac{\text{Estimated disparity} - \text{True disparity}}{\text{True disparity}} \tag{1}$$

We calculated disparity across three different treatments of the ordinated matrices. First, we used the untreated ordination output to examine how well ancestral state estimation recovers the true disparity signal across all PC axes. Second, we min-max standardised the PC axes so that the absolute magnitude in PC scores did not influence the disparity metric scores (i.e.

in this case, the max-score for PC1 was always standardised to 1). Third, we only used the minimum number of dimensions that were contained across all fossil sampling levels for that specific phylogeny so that each fossil sampling level, i.e., the first 48, 98 and 148 dimensions for the 50, 100 and 150 living taxa simulations, respectively. This was to negate any effect that higher-dimensional trait space estimations were more likely to be closer the true disparity signal.

3 Results

3.1 Discrete traits

The relative discrete trait disparity error varied across transition rate, fossil sampling proportion, method of ancestral state estimation and disparity metric.

3.1.1 Effect of estimation method

Pre-ordination estimation (pre-OASE) methods were generally more accurate at recovering the true disparity value than post-ordination estimation (post-OASE) methods. In particular, the pre-OASE distribution method was the best performing ancestral state estimation method.

3.1.2 Effect of estimation method

For both discrete and continuous traits, pre-ordination estimation (pre-OASE) methods were generally more accurate at recovering the true disparity value than post-ordination estimation (post-OASE) methods. The pre-ordination estimation method performs bst. The relative disparity error varied across transition rate, fossil sampling proportion, method of ancestral state estimation and disparity metric. Firstly, we found that the pre-OASE distribution method was the most accurate estimation method for returning the true disparity value across all transition rates, fossil sampling levels and disparity metrics. Furthermore, estimation error decreased as fossil sampling increased. In general, disparity values resulting from slow transition rates were the easiest to recover.

Density metrics, such as the sum of variances and average nearest neighbour distance, tended to be the easiest to recover the true value, whereas size-based metrics were the hardest.

3.1.3 Discrete

3.2 Overall Performance of ASE Methods

4 Discussion

5 Conclusions

Acknowledgments

Author Contributions

Data Availability

Competing Interests

Figures

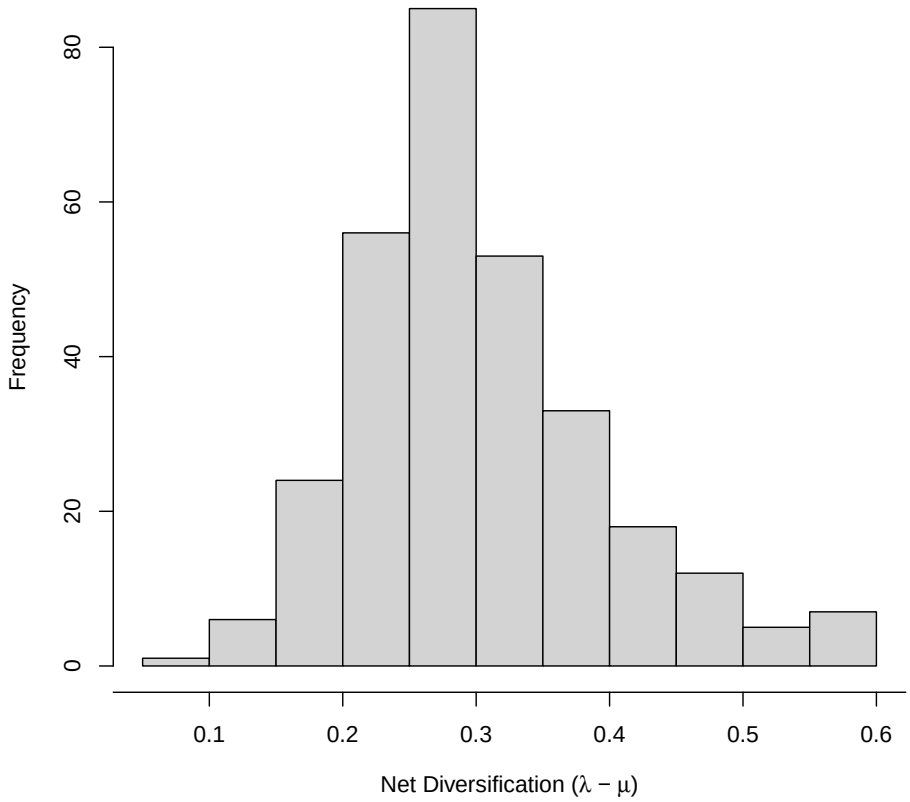

Figure 1: **Birth-death parameters for tree simulation (overall disparity)** Distribution of net diversification values for all trees simulated.

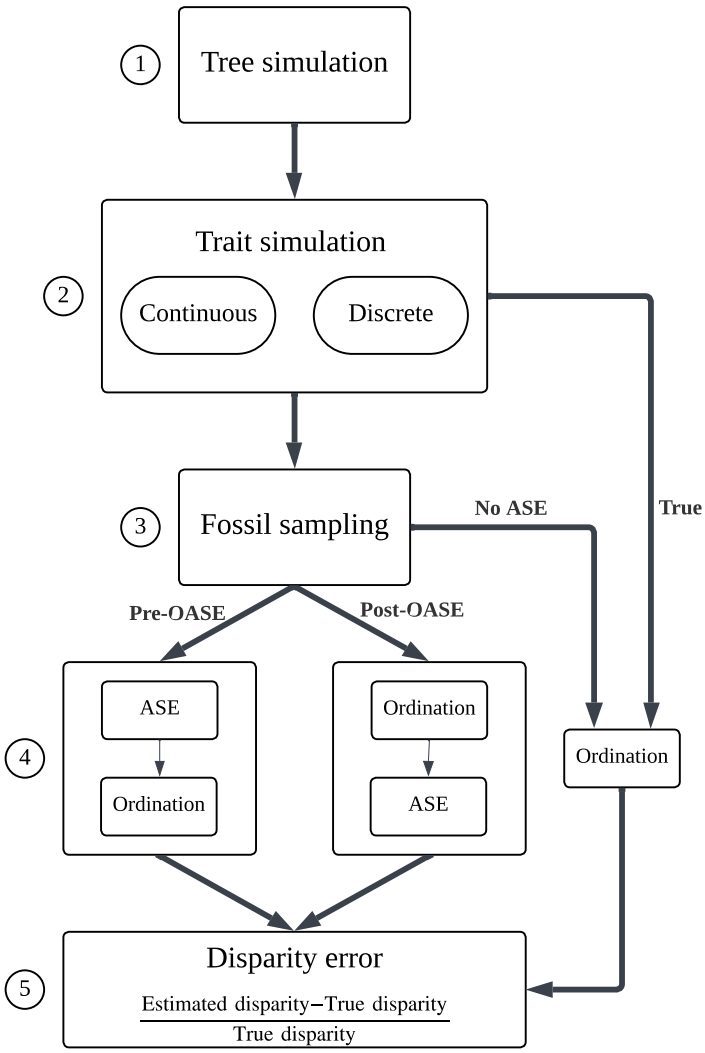

Figure 2: **Simulation workflow**

Tables

Table 1: Trait simulation models and parameters used across each phylogeny. For continuous characters, the root state was always set as 0. P_{ij} = transition probability; σ = variance; α = rate of adaptation toward the optimum; θ = optimum.

Type of data	**Model**	**Parameters**
Discrete	$Fast$	$P_{ij} = 0.9$
	$Medium$	$P_{ij} = 0.1$
	$Slow$	$P_{ij} = 0.01$
Continuous	BM	$\sigma = 0.25$
	BM_t	$\sigma = 0.25,\ \text{trend} = 0.3$
	OU_{st}	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.25}$
	OU_w	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.75}$
	OU_{sh}	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.75},\ \theta = 2$

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